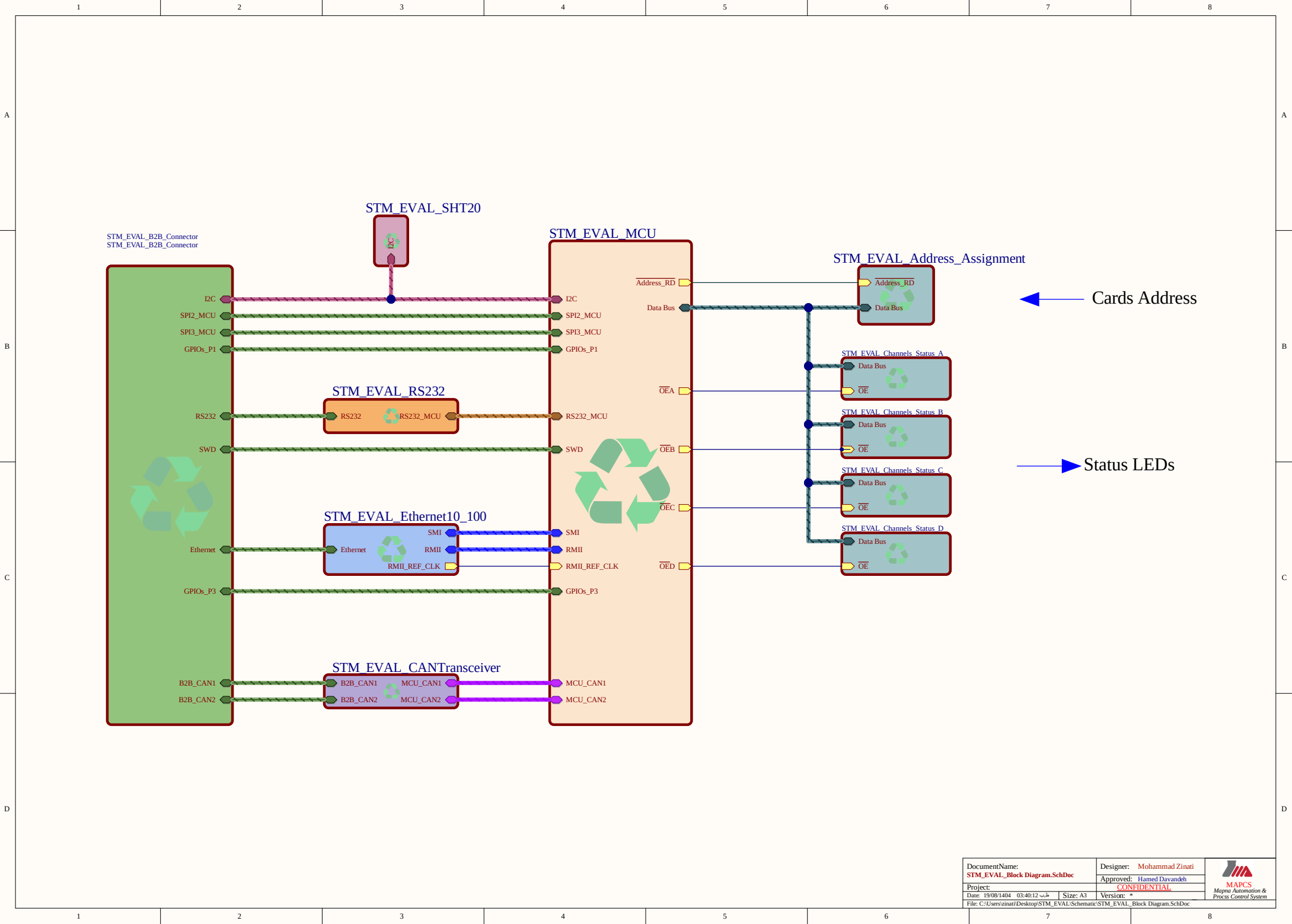




A

B

C

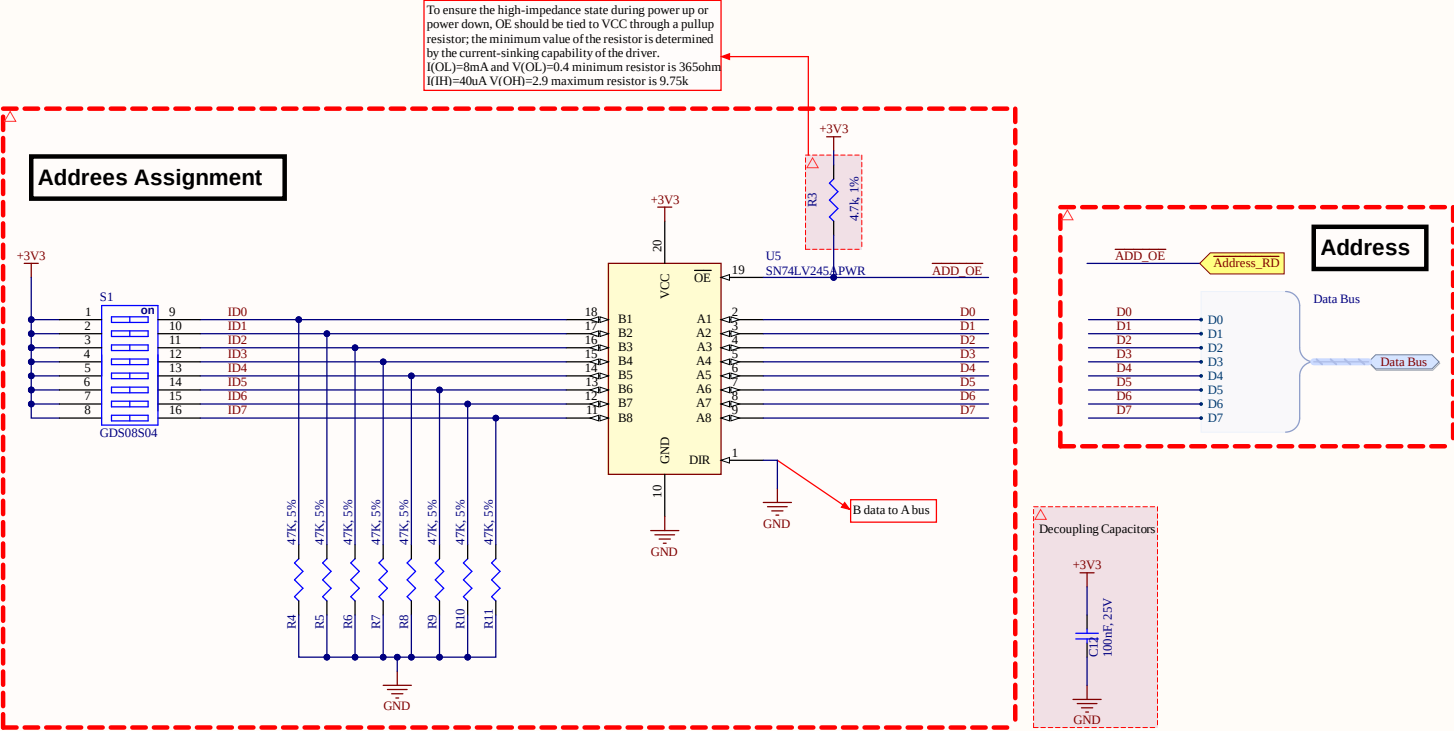
D

A

B

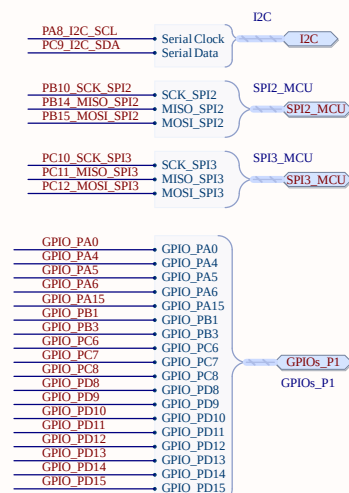
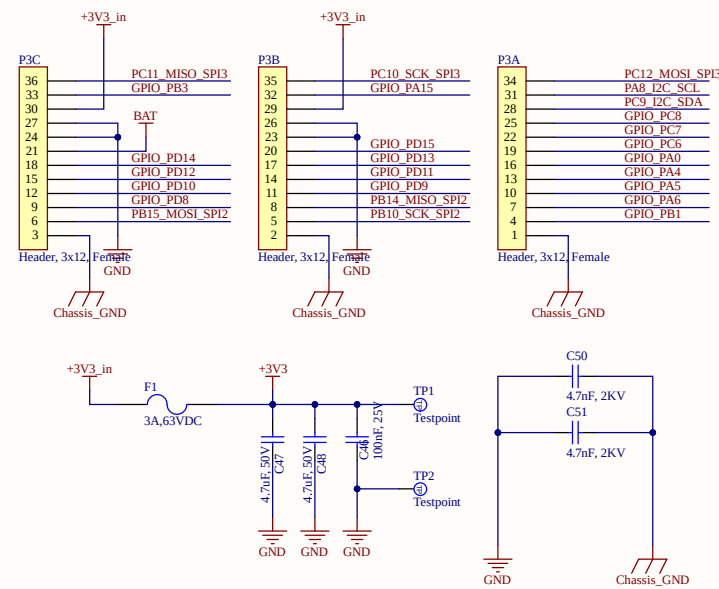
C

D



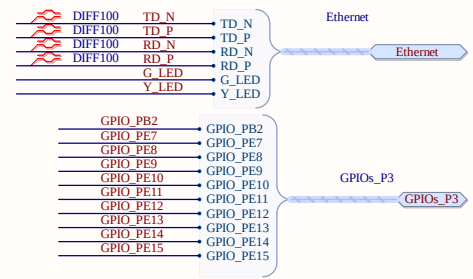
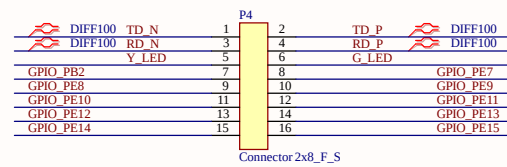
Power & I2C & SPI & GPIOs

P3B
Header, 3x12, Female

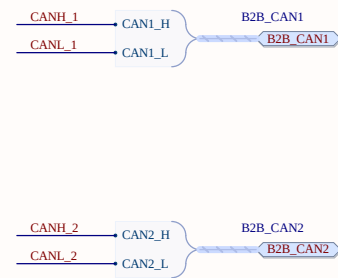
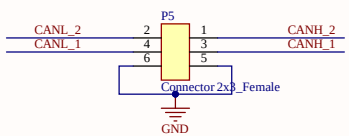


Ethernet & GPIOs

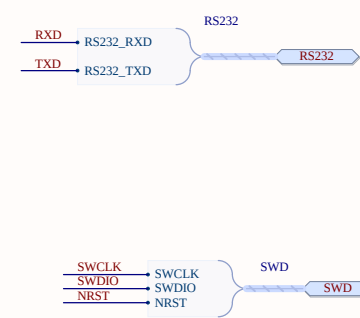
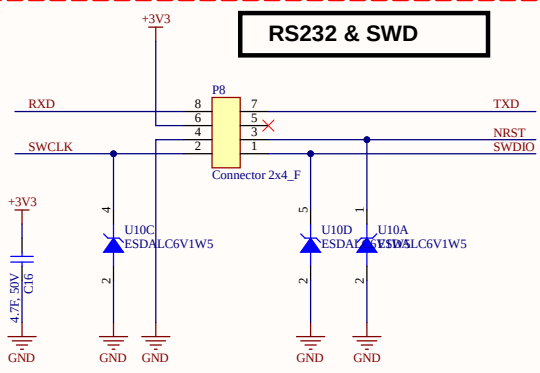
PCB
P700C_2427

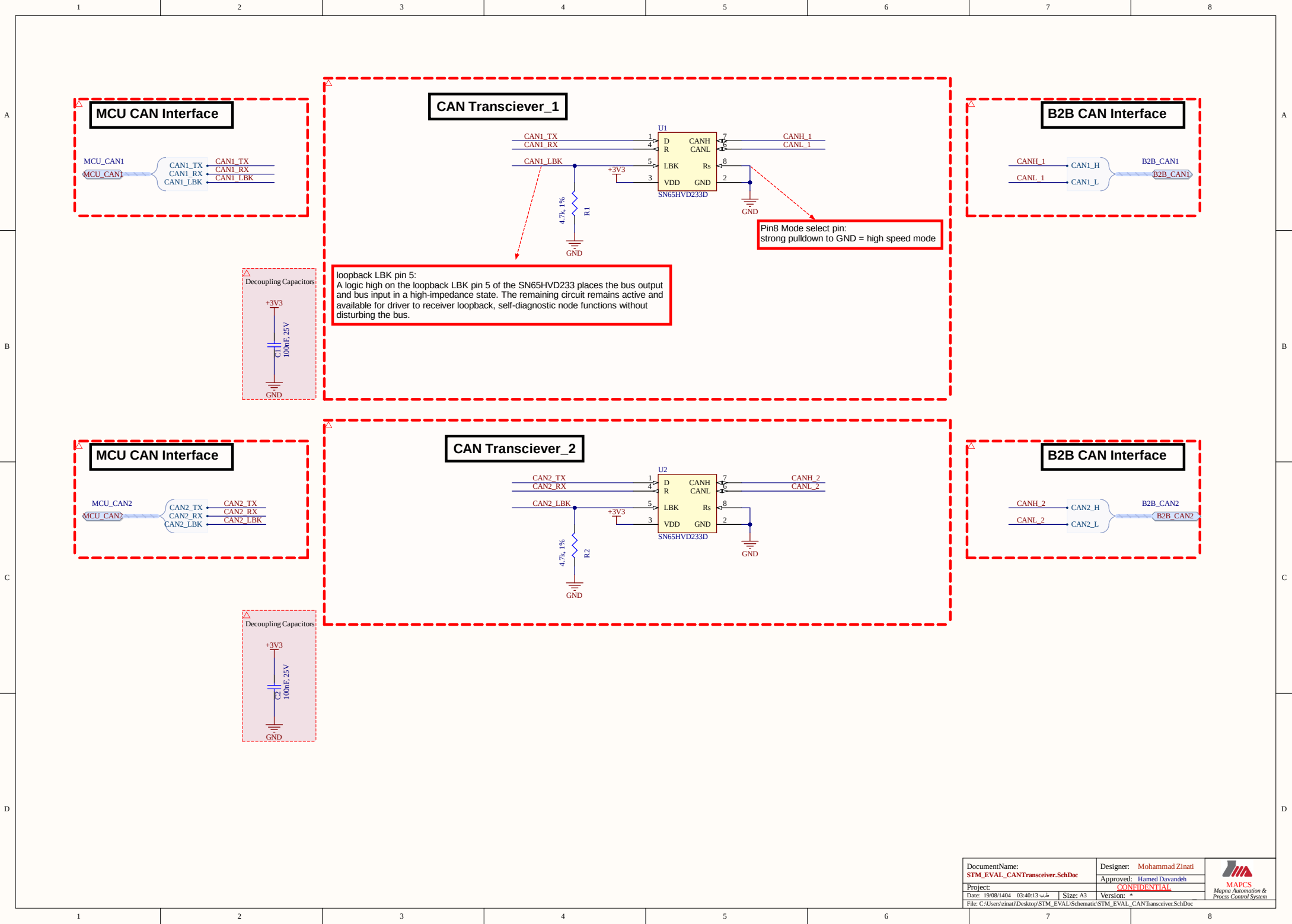


CAN1 & CAN2



RS232 & SWD





A

B

C

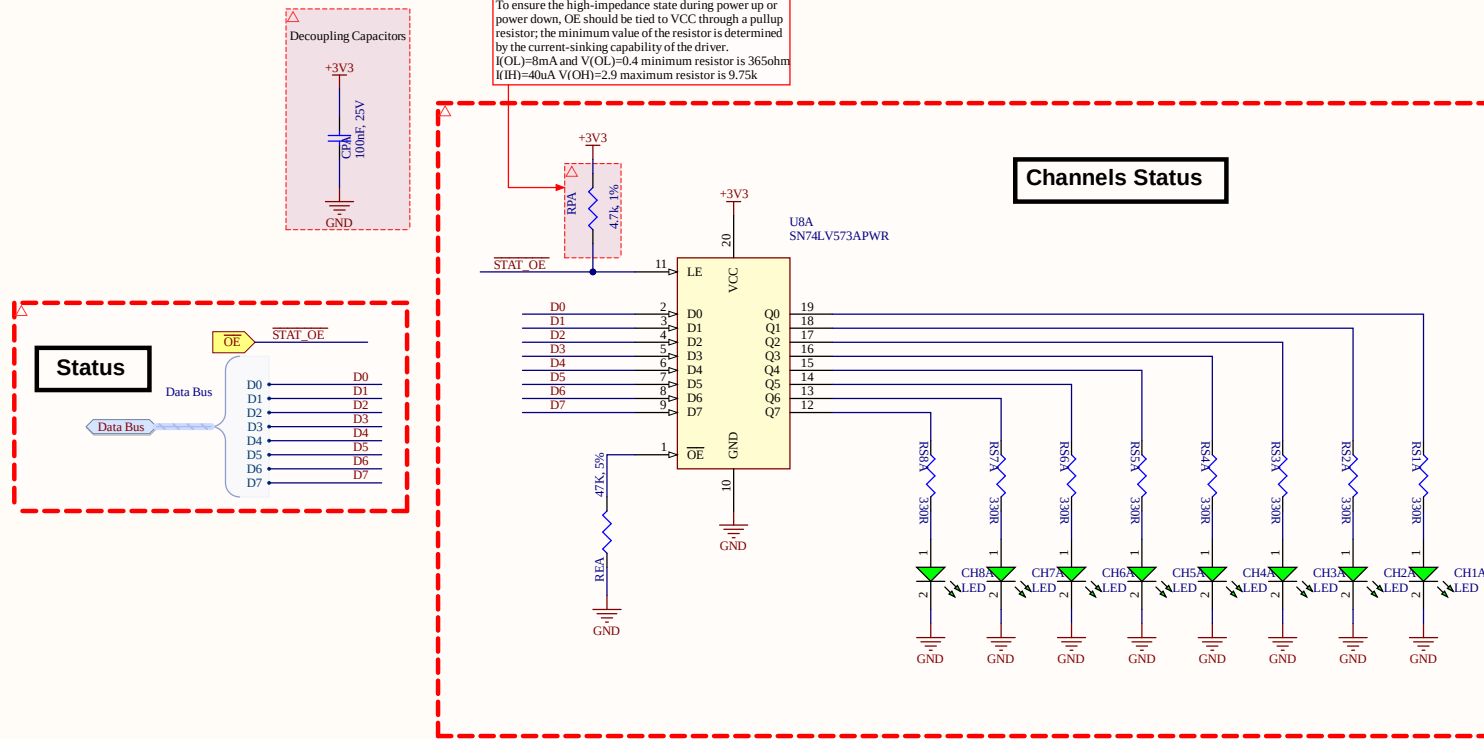
D

A

B

C

D



A

B

C

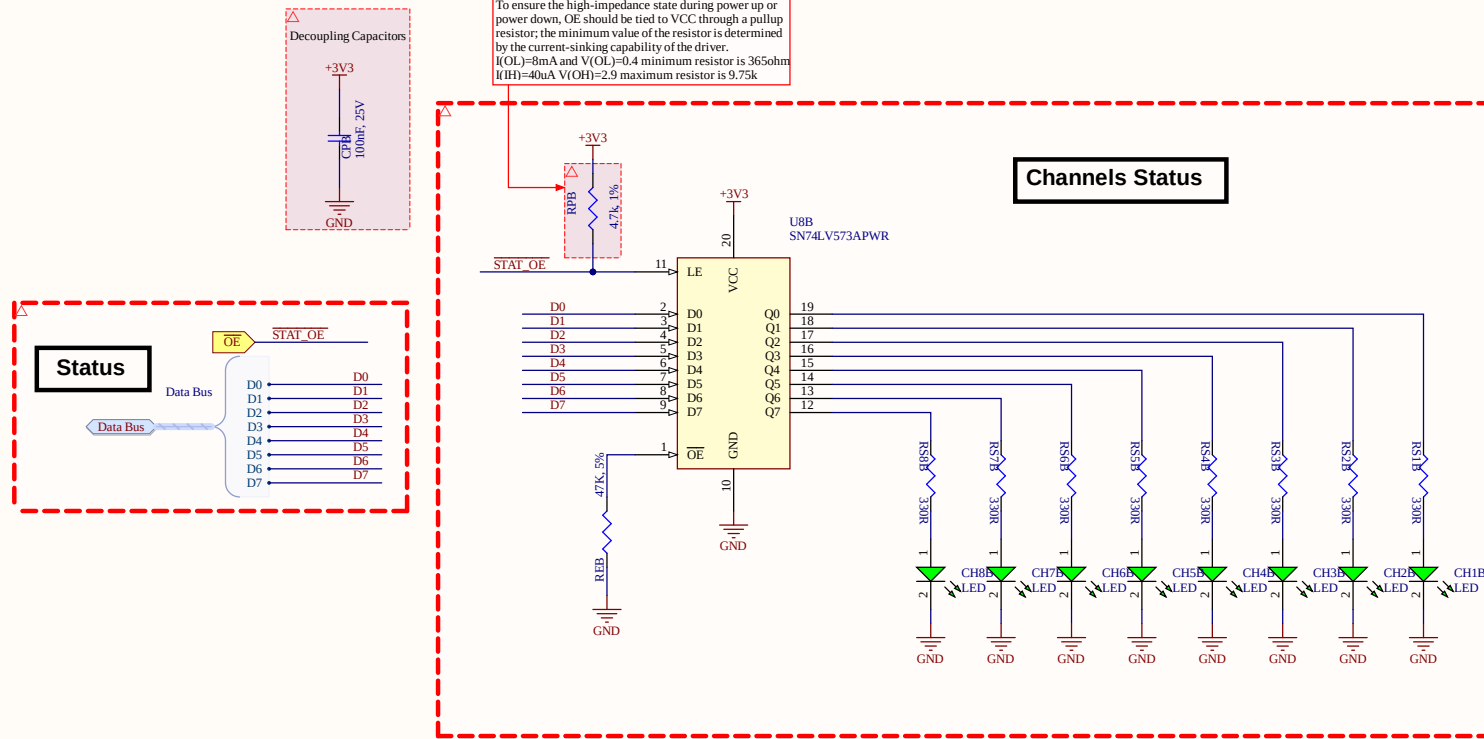
D

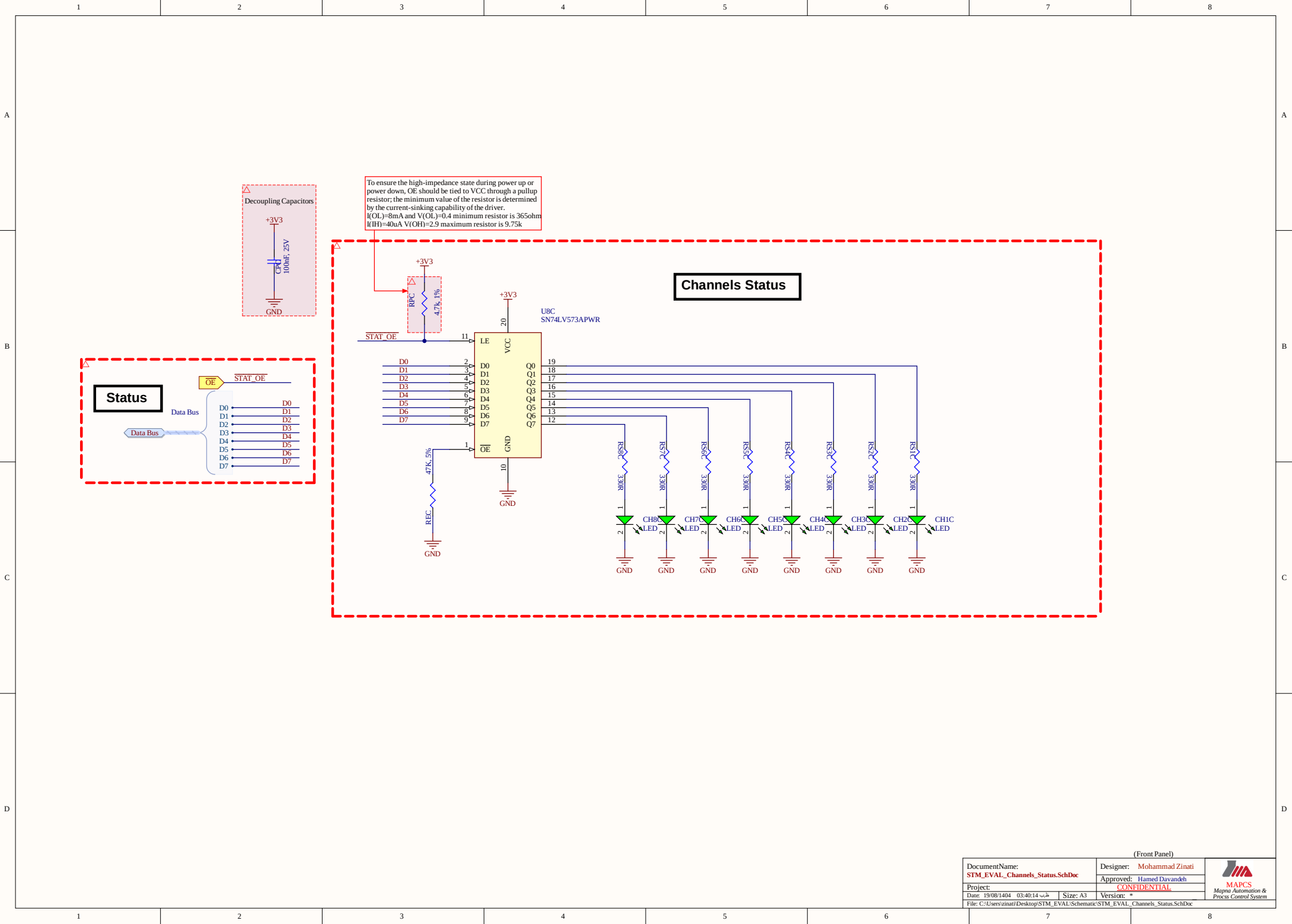
A

B

C

D





A

B

C

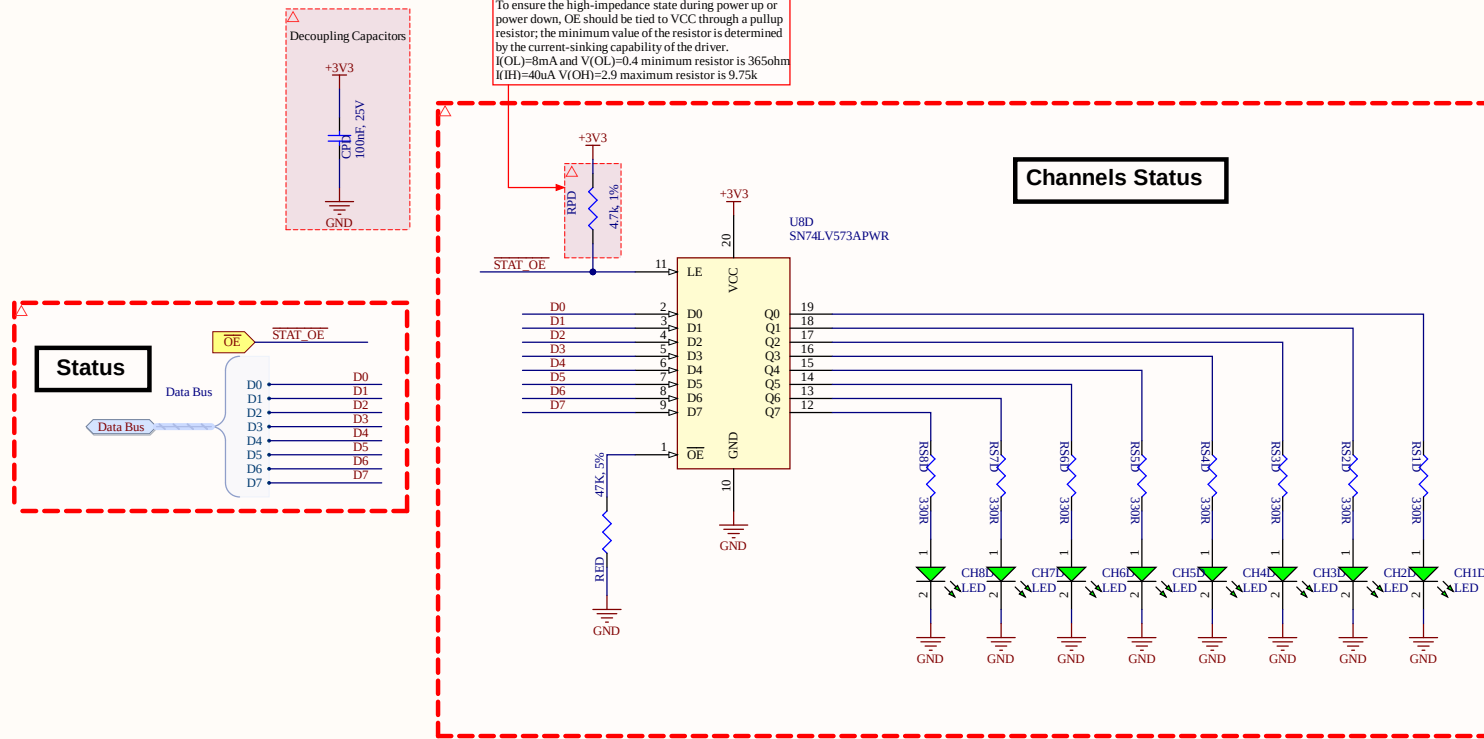
D

A

B

C

D



A

A

B

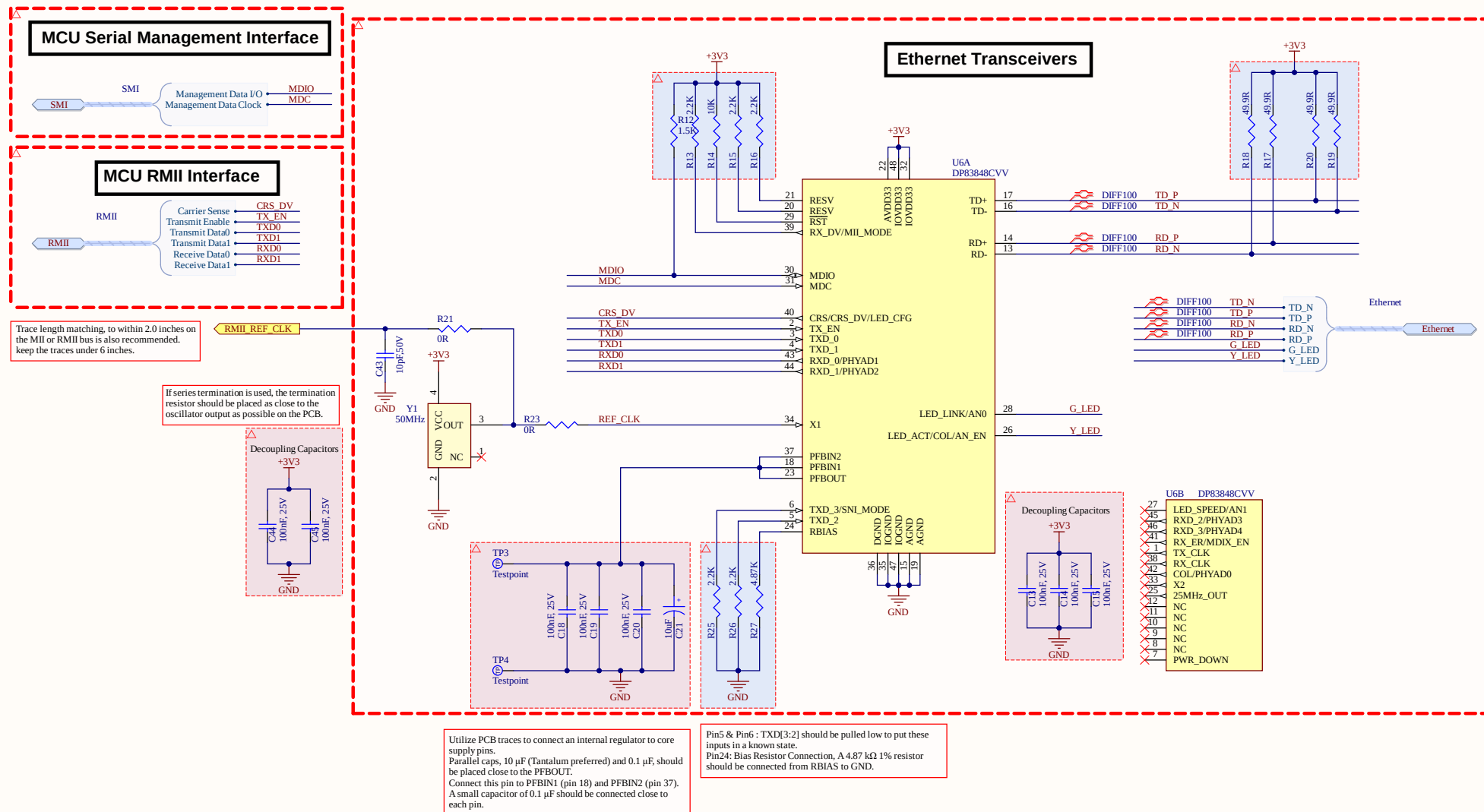
B

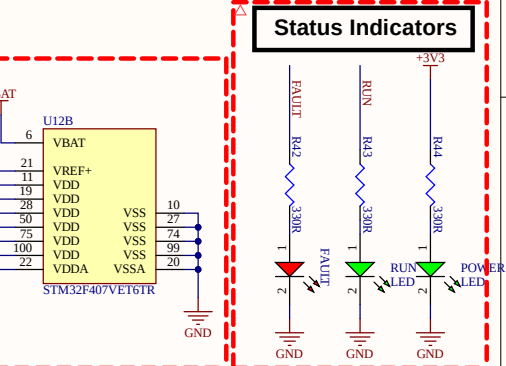
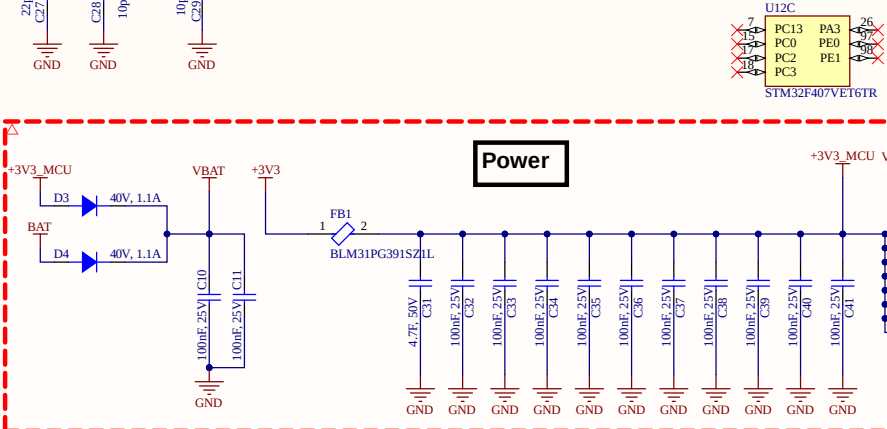
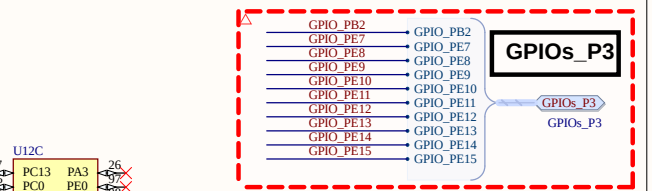
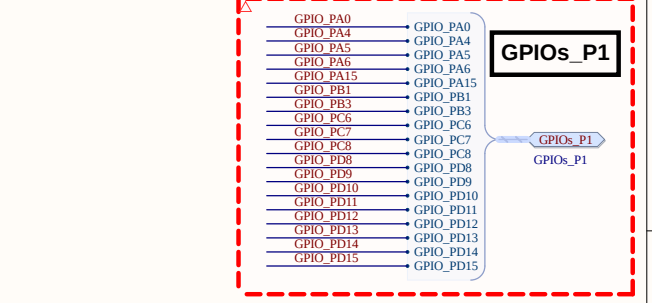
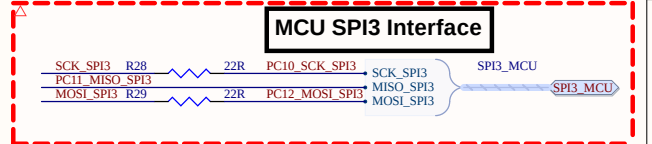
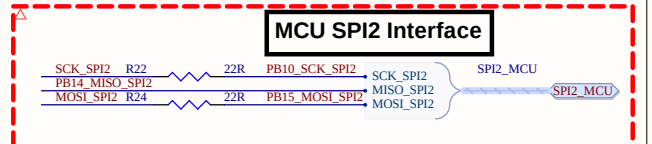
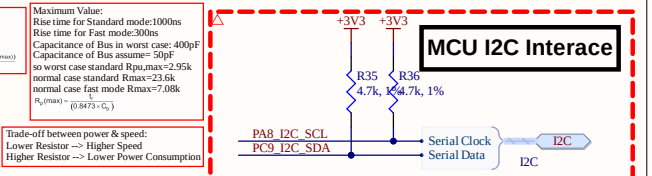
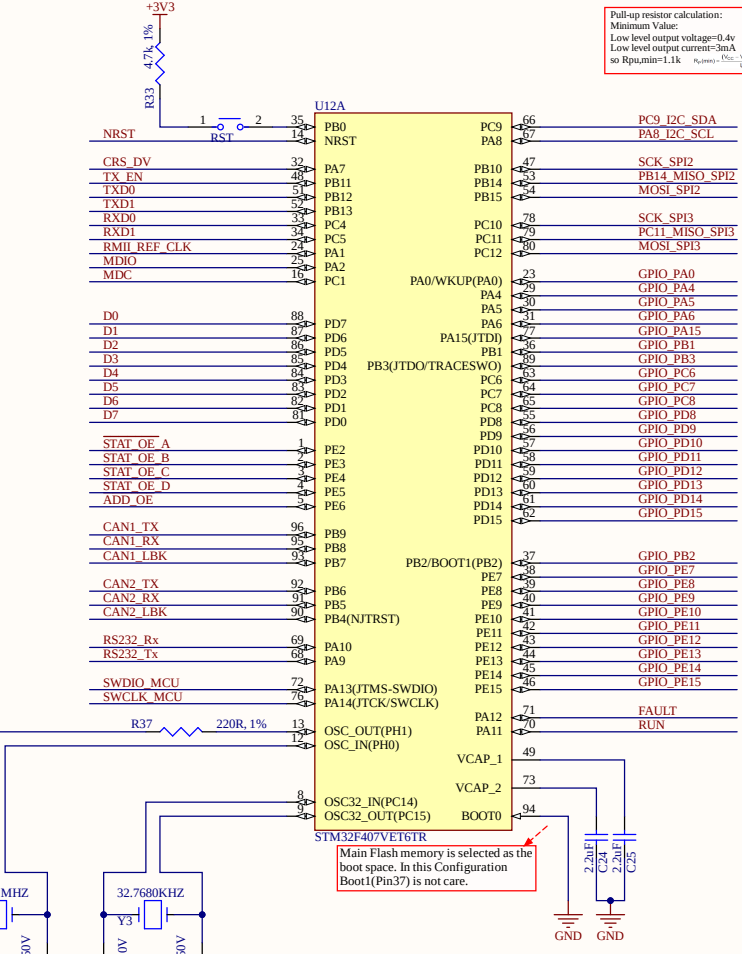
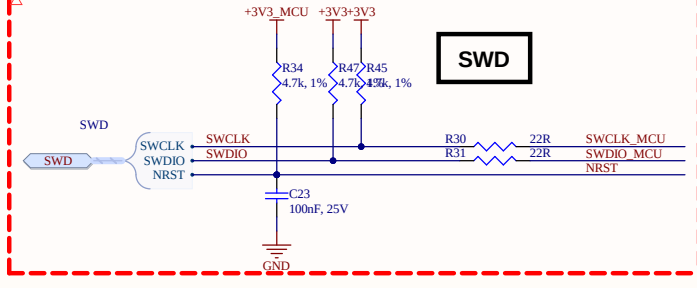
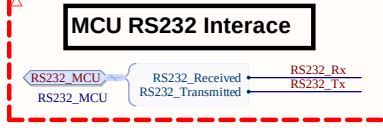
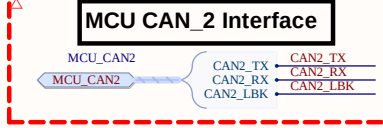
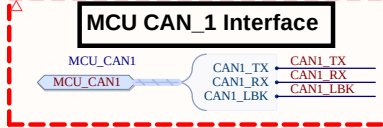
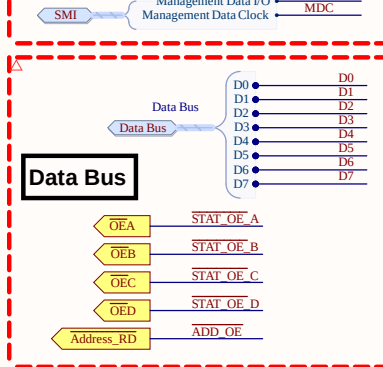
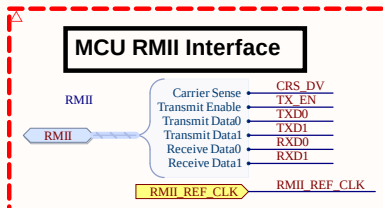
C

C

D

D





The role of this resistor is to limit the drive level of the crystal. With CL2, it forms a low-pass filter that forces the oscillator to start at the fundamental frequency and not at overtones (prevents the oscillator from vibrating at the odd harmonics of the fundamental frequency).

The external capacitors CL1 and CL2 are used to tune the desired value of CL, to reach the value specified by the crystal manufacturer.

$$C_L = \frac{C_1 \cdot C_2}{C_1 + C_2} + C_0$$

Crystal 8MHz:
CL=18pF, Cs=7pF so CL1=CL2=22pF

Crystal 32.768KHz:
CL=7pF, Cs=1.22pF so CL1=CL2=10pF

Main Flash memory is selected as the boot space. In this Configuration Boot1(Pin37) is not care.

Pull-up resistor calculation:
Minimum Value:
Low level output voltage=0.4v
Low level output current=3mA
so $R_{pu,min}=1.1k$

Maximum Value:
Rise time for Standard mode:1000ns
Rise time for Fast mode:300ns
Capacitance of Bus in worst case: 400pF
Capacitance of Bus assume= 50pF
so worst case standard $R_{pu,max}=2.95k$
normal case standard $R_{pu,max}=23.6k$
normal case fast mode $R_{pu,max}=7.08k$

Trade-off between power & speed:
Lower Resistor -> Higher Speed
Higher Resistor -> Lower Power Consumption

A

B

C

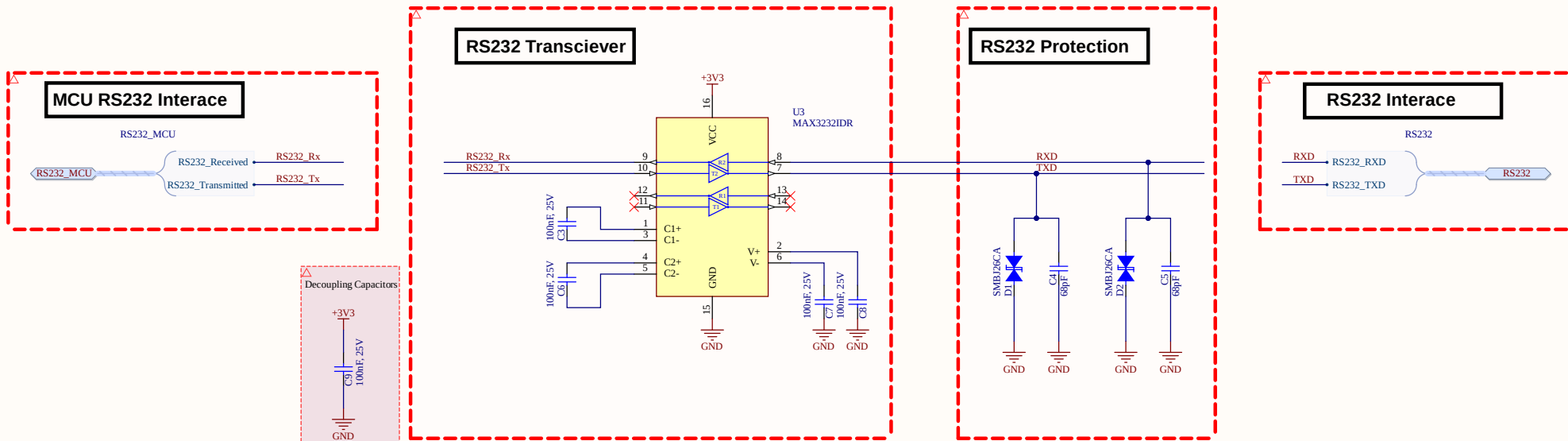
D

A

B

C

D



A

B

C

D

A

B

C

D

